

BARTLESVILLE, OK, USAWMO: **723565**Lat: **36.767N**Lon: **96.017W**Elev: **715**StdP: **14.32**Time Zone: **-6.00 (NAC)**Period: **99-19**WBAN: **03959****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	10.0	14.6	0.1	5.7	14.7	4.7	7.2	19.8	23.0	47.0	20.1	47.8	4.0	0	0.464

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	22.7	100.0	76.1	97.0	76.7	93.5	76.3	80.1	92.5	78.8	91.4	77.8	90.1	9.4	200

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
76.7	143.0	86.9	75.3	136.0	85.7	74.1	130.8	84.2	44.1	92.4	42.8	91.3	41.6	90.2	87.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4)	20.3	18.2	16.3	DB	-0.7	103.4	8.4	4.3	-6.7	106.5	-11.6	109.0	-16.3	111.4	-22.4	114.5
(5)				WB	-1.3	82.8	8.2	1.6	-7.2	83.9	-12.0	84.9	-16.6	85.8	-22.6	87.0

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	58.8	35.5	39.1	49.6	59.1	67.5	76.3	80.6	79.7	71.7	59.6	47.9	37.6
	DBStd	18.11	10.28	11.08	10.56	8.92	7.65	5.21	5.11	5.77	7.09	9.09	9.68	9.82
	HDD50	1524	461	334	135	23	1	0	0	0	0	21	151	398
	HDD65	4021	916	725	487	218	66	2	0	0	21	218	518	850
	CDD50	4733	10	29	123	296	545	790	950	921	650	318	88	13
	CDD65	1755	0	1	10	41	145	341	485	457	221	49	5	0
	CDH74	20067	2	18	128	500	1449	3712	5841	5342	2440	580	54	1
	CDH80	9620	0	2	23	122	462	1645	3169	2923	1105	164	5	0
Wind	WSAvg	6.2	6.3	6.8	7.6	8.0	6.6	6.1	5.0	4.8	5.0	5.6	6.3	6.1
Precipitation	PrecAvg	37.4	1.5	1.7	3.1	3.6	4.7	4.4	3.0	3.1	4.2	3.3	2.7	1.9
	PrecMax	58.6	4.2	6.9	8.8	9.6	10.3	15.0	7.4	10.7	14.6	7.6	6.6	5.7
	PrecMin	19.8	0.0	0.0	0.2	0.1	0.8	0.4	0.0	0.1	0.3	0.2	0.0	0.2
	PrecStd	8.4	1.1	1.5	2.1	1.9	2.4	2.8	2.0	2.2	3.0	2.1	2.0	1.5
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	71.7	76.6	83.0	88.7	90.8	97.6	103.4	105.6	99.7	89.5	79.8	70.9
		MCWB	54.0	59.2	64.2	67.2	72.7	77.4	75.7	75.7	74.7	71.2	62.8	58.7
	2%	DB	65.0	70.2	77.6	83.3	88.1	93.5	99.7	100.5	93.3	84.3	74.4	64.1
		MCWB	51.3	55.5	60.9	65.8	72.1	76.4	76.8	75.5	74.7	68.2	60.3	53.6
	5%	DB	59.7	64.4	72.9	79.8	85.3	91.1	96.8	97.4	90.3	80.7	70.2	59.6
		MCWB	48.8	53.5	58.8	63.8	71.3	76.1	77.2	76.5	73.6	65.9	58.5	50.5
	10%	DB	53.9	59.1	68.7	75.8	82.1	88.7	93.4	93.6	86.8	76.4	65.1	54.7
		MCWB	44.8	49.0	57.0	62.9	70.1	75.1	77.3	75.9	71.9	64.4	55.0	47.4
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	59.8	62.8	67.2	71.3	77.0	80.4	82.0	81.8	78.2	74.4	66.8	63.1
		MCDB	64.7	71.4	79.0	82.0	86.8	92.3	93.6	93.4	90.7	84.4	75.3	68.1
	2%	WB	54.0	59.0	64.3	69.0	74.9	78.9	80.2	79.8	76.5	71.9	63.5	57.0
		MCDB	61.4	66.7	73.2	78.7	84.3	90.5	92.6	92.8	88.7	80.0	70.0	61.6
	5%	WB	49.3	54.7	61.7	67.2	73.3	77.6	79.0	78.3	75.2	69.7	60.5	52.2
		MCDB	59.2	63.0	69.9	76.0	82.3	88.6	91.7	91.5	86.8	76.1	67.2	57.0
	10%	WB	45.3	49.2	58.6	65.0	71.5	76.2	78.0	77.1	73.6	66.6	57.0	47.7
		MCDB	53.1	58.5	67.1	73.5	79.9	86.5	90.5	90.0	83.9	73.5	64.1	54.2
Mean Daily Temperature Range	5% DB	MDBR	24.4	24.3	25.2	24.5	22.1	21.2	22.7	24.2	25.4	25.6	25.4	23.1
		MCDBR	34.9	33.4	31.7	29.5	25.4	23.8	27.1	29.3	28.7	29.8	31.2	31.7
		MCWBR	22.5	21.0	17.5	15.5	12.5	10.2	9.1	9.6	11.9	15.3	18.5	21.4
	5% WB	MCDBR	31.0	29.2	24.1	23.9	21.4	21.0	22.7	24.5	24.4	22.0	24.6	25.1
		MCWBR	21.7	19.9	15.1	14.7	11.8	10.1	9.5	9.9	10.8	13.2	16.9	20.2
Clear-Sky Solar Irradiance	taub	0.298	0.310	0.354	0.386	0.423	0.434	0.444	0.446	0.394	0.350	0.313	0.293	
	taud	2.485	2.465	2.378	2.313	2.269	2.267	2.284	2.241	2.363	2.467	2.508	2.515	
	Ebn at Noon	285	295	289	284	274	270	266	263	272	276	277	280	
	Edh at Noon	26	30	36	40	43	43	42	43	36	30	25	23	
All-Sky Solar Radiation	RadAvg	801	1003	1313	1648	1829	2095	2092	1884	1578	1173	855	680	
	RadStd	77	152	78	107	172	185	127	125	126	140	99	74	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (35 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A